

Step 2 of **R2B (X)** – Let's normalize the following binary numbers!

$$\begin{aligned} 1. \quad 101011010.101_2 &= 101011010.101_2 \times 2^0 \\ &= 1.01011010101_2 \times 2^8 \end{aligned}$$

$$\begin{aligned} 2. \quad 0.000000001101_2 &= 0.000000001101_2 \times 2^0 \\ &= 1.101_2 \times 2^{-9} \end{aligned}$$

$$\begin{aligned} 3. \quad -11000000111001_2 &= -11000000111001._2 \times 2^0 \\ &= -1.1000000111001_2 \times 2^{13} \end{aligned}$$

B2R (X) – Reversing the normalization, i.e.,
denormalized the following binary #'s (**V**)

$$\begin{aligned} 1. \quad -1.111101_2 \times 2^5 &= -111110.1_2 \times 2^0 \\ &= -111110.1_2 \end{aligned}$$

$$\begin{aligned} 2. \quad 1.0_2 \times 2^{-8} &= 0.000000010_2 \times 2^0 \\ &= 0.00000001_2 \end{aligned}$$

$$\begin{aligned} 3. \quad 1.1000000111001_2 \times 2^{-1} &= 0.11000000111001_2 \times 2^0 \\ &= 0.11000000111001_2 \end{aligned}$$

Let's practice all 3 steps of **B2R (X)** ! (single precision)

Which real number does **0xBF950000** (**B**) represent on our *target machine*?

1. **0xBF950000** ->

1	01111111	001010100000000000000000
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Determine ...

s = 1 -> $(-1)^1$ -> negative number

E = **exp** - **bias** = $B2U(01111111_2) - 127 = 127 - 127 = 0$

M = 1. + **frac** = $1. + 001010100000000000000000_2 = 1.0010101_2$

Binary numerical form: **V** = $(-1)^1 1.0010101_2 \times 2^0$

V = $(-1)^s \mathbf{M} 2^E$

2. Denormalize **V**: -1.0010101_2

3. Using the **positional notation**, convert the above denormalized binary number into **R**: $-1_2 = -1$ $.0010101_2 = 1/8 + 1/32 + 1/128$

Answer: -1.1640625

$= 0.125 + 0.03125 + .0078125 = 0.1640625$

Homework

Let's practice all 3 steps of **R2B (X)** ! (single precision)

How would **12345.75** be encoded as IEEE 754 floating point number?

1. Convert 12345.75 to binary

$$\Rightarrow 12345 \Rightarrow 11000000111001_2 \quad .75 \Rightarrow .11_2 = 11000000111001.11_2$$

2. Normalize binary number: $11000000111001.11_2 \times 2^0 = 1.100000011100111_2 \times 2^{13}$

3. Determine ...

Binary numerical form: $V = (-1)^s M 2^E$

$s = 0 \rightarrow (-1)^0 \rightarrow$ positive number

$E = \text{exp} - \text{bias}$ where $\text{bias} = 2^{k-1} - 1 = 2^7 - 1 = 128 - 1 = 127$

$\text{exp} = E + \text{bias} = 13 + 127 = 140 \Rightarrow \text{U2B}(140) = 10001100_2$

$M = 1. + \text{frac} \rightarrow \text{frac} = M - 1. = 1.100000011100111_2 - 1. = 100000011100111_2$



In **hex**: **0x4640E700**